Blockchain-Based Decentralized and Lightweight Anonymous Authentication for Federated Learning

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***Abstract*—Federated learning (FL) is a promising technology for achieving privacy-preserving edge intelligence and has attracted extensive attention from industry and academia. However, in the FL training process, the server directly aggregates local models from mobile devices, which poses serious privacy and security threats. The identity authentication mechanism can provide FL with local model integrity and source authentication. However, the existing schemes are centralized, and most of them are computationally expensive, resulting in limited performance. To address these issues, this paper proposes a decentralized and lightweight anonymous FL identity authentication scheme, namely DAFL. In our scheme, we first design a decentralized and simplified storage FL authentication framework by combining the directed acyclic graph (DAG) blockchain and accumulator. Then, we propose a lightweight digital signature algorithm that supports batch verification for authentication. Finally, nodes interact through pseudonyms to achieve anonymous communication, and the trusted authority (TA) can track and recover the real identities of nodes when malicious behavior occurs. We theoretically prove the security of the proposed DAFL. The extensive experiments demonstrate that DAFL achieves lower authentication overhead and better convergence performance compared to existing authentication schemes and vanilla FL systems.**

***Index Terms*—Blockchain, federated learning, authentication, batch verification.**

I. Introduction

With the increasing industrialization of the Internet of Things (IoT) in multiple real-world applications such as smart transportation and smart city, various sensor devices are producing massive amounts of sensing data. How to obtain and analyze useful information from a large amount of data has attracted extensive attention. The rise of artificial intelligence (AI) can significantly promote Big Data analysis in industrial IoT. As a powerful analytical tool, AI can provide accurate and efficient data analysis. Massive data can be utilized by machine learning technologies to perform various analyses and provide useful information for decision-making analysis, prediction and classification of future mobile device behavior [1]. However, machine learning needs to collect user data, which can result in data privacy leakage.

Several techniques have been proposed to mitigate the risk of privacy breaches. One solution is covert communication, which hides the existence of wireless transmissions [2], [3]. Another way is to use blockchain technology to securely share data and select reliable devices [4], [5]. Federated Learning is a new paradigm to train AI models while preserving user privacy, by keeping data local and sharing only trained models [6], [7], [8], [9]. However, FL has limitations, such as the lack of participant identity authentication, which can cause issues like invalid or malicious shared models. Firstly, participants may disavow their shared local model, which may be invalid or even malicious. Secondly, attackers may compromise legitimate participants and share poisoned local model parameters with the server. This can lead to inaccurate global model aggregation, distorting local model updates at the participants. Therefore, it is necessary to perform identity authentication for participants to conduct FL safely and efficiently.

The mainstream identity authentication solution is the Public Key Infrastructure (PKI) technology, which can achieve large-scale identity authentication. The PKI relies on a Certificate Authority (CA). The CA, as a trusted centre, leverages its own private key to sign and issue digital certificates for users or other CAs to bind users' public keys and identities [10]. That is, PKI can endorse the identity of the public key holder to ensure that users can authenticate each other. Zhao et al. [11] proposed a FL authentication scheme for smart transportation systems leveraging PKI. CA generates a certificate for each vehicle according to their privacy information. After receiving the local model parameters of the vehicle, other nodes check whether the vehicle's certificate matches its identity information, so as to realize the authentication between vehicles, RSUs, and servers in the FL process. However, some shortcomings exist in PKI authentication [12]. Firstly, the centralized feature of

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PKI can cause the single point of failure. Once the private key of a CA is leaked, all nodes must discard the corresponding certificate of the CA, and attackers can easily carry out man-in-the-middle attacks, thus damaging the security of users. Secondly, it is difficult for users to manage the certificates of their devices. Thirdly, device manufacturers often generate a key for the device and install it directly. Therefore, the manufacturer may leak the secret key or become the target of attackers.

In order to remedy the existing problems of PKI authentication, Identity-based Cryptography (IBC) [13] is proposed, which directly leverages the effective identity of the entity as the public key, making the authentication process no longer limited to the certificate mechanism, and simplifying the management process of the key corresponding to the entity identity. Wang et al. [14] proposed an IBC-based FL identity authentication scheme for electric vehicles, which implements identity authentication during training through a digital signature algorithm. However, although the IBC authentication scheme does not store and manage certificates, it still needs to manage the information related to public keys, and the private keys are generated by Key Generation Center (KGC), which has problems with key management, centralization and only applicable to small-scale trusted domain networks.

Existing FL authentication schemes are centralized and cannot guarantee the identity privacy of nodes. Furthermore, existing schemes lead to a large amount of computing overhead, which limits their performance. In order to solve these problems, we propose a decentralized and lightweight anonymous authentication scheme for FL, namely, DAFL. DAFL eliminates the centralized KGC, meets the security requirements and supports batch authentication of messages. The primary contributions of this paper are summarized as follows:

1) We propose a decentralized and efficient FL authentication framework, where nodes generate their own public-private key pair and record the identity and public key in the blockchain accumulator, thus solving identity storage and key management problems.
2) DAFL realizes anonymous communication of nodes, which can not only ensure the identity privacy of nodes but also track the real identity of malicious nodes.
3) We propose a lightweight digital signature algorithm for FL authentication, which does not use computationally heavy bilinear pairings and reduces the computational overhead. In addition, the efficiency of FL authentication is improved by allowing the receiver to batch authenticate multiple received messages.
4) Theoretical analysis proves the unforgeability of the proposed signature algorithm and verifies the security of DAFL. The comprehensive comparison and performance analysis with the existing schemes show that DAFL has conspicuous advantages in many aspects.

The rest of this paper is organized as follows. Section II discusses the related work of identity authentication. Section III reviews the preliminaries, and Section IV introduces the overview of DAFL. The implementation details of the FL decentralized anonymous identity authentication scheme are presented in Section V. Section VI describes the detailed security analysis, and Section VII conducts a comprehensive comparison and performance analysis. Finally, Section VIII concludes the paper.

II. Related Work

PKI Authentication Schemes: The PKI has been widely studied as the infrastructure of identity authentication. The traditional PKI is established based on the CA. Through the digital certificate issued by the CA, the user's public key and identity are bound. Siddiqui et al. [15] proposed a PKI digital certificate authentication protocol for cloud IoT systems, which is based on two CAs. In order to prevent malicious vehicles accessing VANET, Azees [16] and Vijayakumar [17] proposed PKI-based anonymous authentication schemes, and both of them provided a tracking function to track vehicles that compromise VANET. Tao et al. [18] proposed an identity authentication scheme combining PKI and combined public key in the context of mobile crowd sensing. However, due to the centralized nature of PKI, a single point of failure or a clear target may affect the entire authentication mechanism.

PKI authentication schemes based on blockchain: Several recent works [19], [20], [21] leverage blockchain to overcome the limitations of traditional PKI. Kubilay [20] and Qin et al. [21] successively proposed decentralized PKI systems based on the blockchain with decentralized characteristics. By recording the certificate on the blockchain, PKI's certificate management authority is delegated to the device owner so that they can manage the certificate themselves. Feng et al. [22] proposed an authentication service model to provide one-stop identity authentication services for third-party service providers through the certificate blockchain. Subsequently, Ma et al. [23] proposed a decentralized PKI identity authentication and key management mechanism, which was implemented based on blockchain and bivariate polynomials. Considering the limited storage capacity of lightweight devices and the inability to download the entire blockchain, Jiang et al. [24] proposed a decentralized PKI authentication scheme for lightweight devices. However, during identity authentication, multiple other nodes need to be combined for verification, and the verification process is complicated and requires a lot of additional communication overhead. Although these schemes alleviate the shortcomings of traditional PKI to some extent, there are still some problems such as the distribution, revocation and verification of public key certificates.

IBC Authentication Schemes: Considering the PKI certificate management problem in the identity authentication process, Shamir [25] first proposed the idea of an identity-based cryptosystem at the American Cryptography Association. It was not until 2001 that Boneh and Franklin [26] designed the first feasible IBC scheme based on Shamir's vision. In this scheme, the public key can be any string related to the user's identity, such as name, or e-mail address, and the relevant private key is securely obtained from a trusted third-party key generation

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centre. Afterwards, many related improvement and extension schemes [27], [28], [29] appeared one after another. Sun et al. [27] proposed an IBC-based authentication scheme that avoids the use of certificates in the authentication process. Wang et al. [28] proposed a two-way authentication scheme implemented by a new group signature-based algorithm. The signature algorithm has zero-knowledge properties, but this paper does not introduce the security proof of the algorithm. Sikarwar et al. [30] proposed an authentication scheme for node dynamic and unevenly distributed environments, which implements batch authentication through bilinear mapping and a one-way hash function. These schemes solve many problems in the implementation of PKI authentication, such as heavy certificate management, complex certificate usage process, and so on. However, although these schemes cancel certificates, they still need to store and manage some information, and there are still information management problems.

IBC authentication schemes based on blockchain: Similarly, some works [31], [32], [33] propose to solve this problem through blockchain. Xu et al. [31] proposed a blockchain-based authentication mechanism, which utilizes the tamper-proof feature of the blockchain to manage the user's public key, thereby ensuring the non-repudiation of signatures. Ren et al. [32] proposed an authentication scheme that utilizes the immutability and openness of the blockchain to manage and verify the legitimacy of device identities. Tong et al. [33] proposed a privacy-preserving authentication scheme based on the consortium blockchain. The blockchain serves as a public ledger to record the public keys of all devices to achieve cross-domain authentication. However, these IBC authentication systems rely on KGC to centrally generate the private key of the device, which has the defects of centralization and limited application scale. In addition, they all use computationally complex bilinear pairings for signature and verification, resulting in high computational overhead for identity authentication.

III. Preliminaries

We review some fundamental preliminaries related to DAFL, including blockchain, standard digital signature and dynamic accumulator.

A. Blockchain Technology

Blockchain is a promising technology combining cryptography, distributed consensus and timestamp, which can collaborate in an untrusted environment. Blockchain has a variety of properties, including decentralization, immutability, transparency, and security. The decentralization function alleviates the situation of low efficiency, untrustworthiness, and privacy leakage in the common centralized environment. However, the single-chain structure of the traditional blockchain greatly limits the performance of blockchain applications, such as low transaction efficiency and throughput [34]. Moreover, once the single-chain structure forks, some honest transactions will be lost and the resources of the blockchain system will be wasted. Considering the above defects, this paper adopts the DAG blockchain structure, which can generate blocks in parallel and improve the scalability of the system.

B. Standard Digital Signature

The formal definition of a digital signature is described below.

1) *Initialization:* System initialization generates the basic parameters $(M, S, K, Sign, Ver)$ of the signature scheme, where M is the message space, S is the signature space, K is the key space, including the private key and the public key, $Sign$ is the signature algorithm set, and Ver is the signature verification algorithm set.
2) *Generate a signature:* For each $k = (k_1, k_2) \in K$, where k_1 is the public key, k_2 is the private key, and there are signature algorithms $sign_{k_2} \in Sign$, $sign_{k_2} : M \to S$. For any message $m \in M$, if $s = sign_{k_2}(m)$ and $s \in S$, then s is the signature of the message, and (m, s) is sent to the signature verifier.
3) *Verify the signature:* For the above $k_1 \in K$, there are corresponding signature verification algorithms $ver_{k_1} : MS \to \{True, False\}$, $ver_{k_1} \in Ver$, and

$$ver_{k_1}(m, s) = \begin{cases} True & \text{if } s = sign_{k_2}(m) \\ False & \text{if } s \neq sign_{k_2}(m). \end{cases}$$

After receiving (m, s), the signature verifier calculates $ver_{k_1}(m, s)$. If $ver_{k_1}(m, s) = True$, the signature is valid, otherwise the signature is invalid.

Definition 1. (Discrete logarithm (DL)): Let E be an elliptic curve over a finite field F_p, let $\mathcal{G}_1 = (E, +)$, where P is a point with prime q on E. Let $G = \langle P \rangle \leq \mathcal{G}_1$, give $aP \in G$, and the coefficient $a \in Z_q^*$ is unknown, and a is the value to be solved.

Definition 2. (DL assumption): If there is no probability polynomial-time adversary $\mathcal{A}$ to solve the DL problem with the advantage of ε, then the DL problem is considered to be a difficult problem.

Definition 3. Algebraic Group Model (AGM) [35]: Modelling as an algebraic adversary $\mathcal{A}$, while outputting the elements in the group, provides a representation of that element based on all previous group elements.

C. Dynamic Accumulator

The accumulator can map a set containing multiple identities and public key information to an accumulator value while producing a short witness that a specific element does indeed belong to the set [36]. In our construction, the dynamic accumulator based on RSA introduced by Boneh [37] is invoked as a primitive. This scheme can be described as a five-tuple (Wit, Ver, Add, Del, $MemWitUp$).

1) $Wit(f_i, Acc_F, F)$*:* Input accumulator value Acc_F, element $f_i \in F$, $i = 1, 2, \ldots, n$ and accumulator set F to generate witness wit_i of f_i.
2) $Ver(f_i, wit_i, Acc_F)$*:* Input the accumulator value Acc_F, element f_i and its witness wit_i. Returns 1 if wit_i is a witness of $f_i \in F$; otherwise, returns 0.

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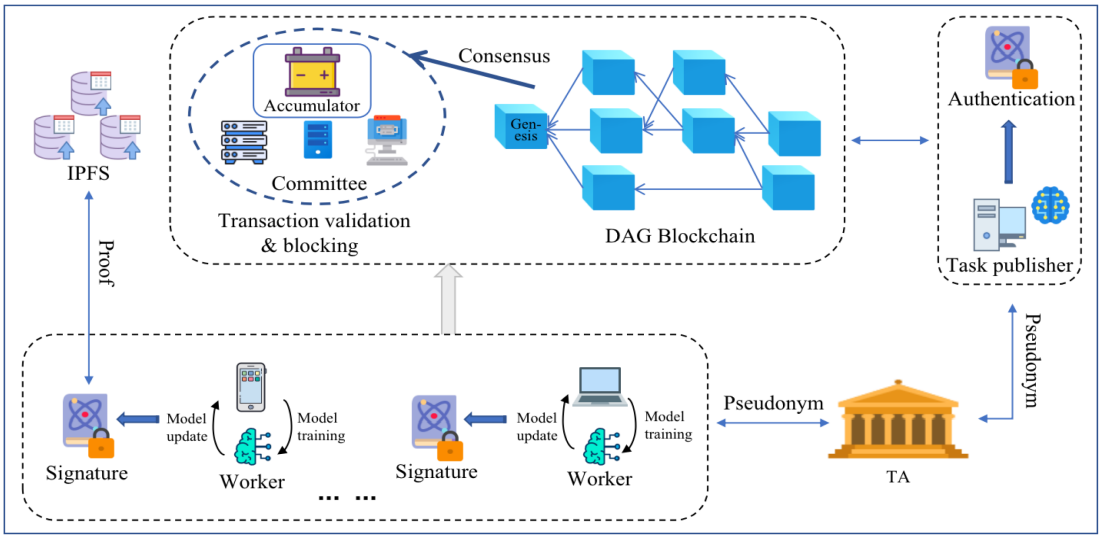


Fig. 1. Framework of the proposed DAFL system.

3) $Add(Acc_F, F, f_i)$: Add element $f_i \notin F$ to the accumulator and update the accumulated value Acc_F, and auxiliary information $upmsg$.
4) $Del(Acc_F, F, f_i)$: Delete element $f_i \in F$ and update the accumulator value $Acc_{F'}$ and auxiliary information $upmsg$.
5) $MemWitUp(Acc_F, f_i, wit_i, upmsg)$: After adding element f_i, the user updates the witness wit_i of element f_i.

The proposed DAFL improves the verification efficiency of identity and public key pair while realizing FL authentication through the accumulator, and solves the problem of insufficient storage space for light nodes when the blockchain volume is growing.

IV. Problem Description

We first outline the system model of DAFL, then introduce the threat model and security requirements of the system. Finally, the design goals are proposed.

A. System Model

We introduce the DAFL system architecture, shown in Fig. 1, which involves four entities, namely trusted authority (TA), task publishers, worker nodes, and the blockchain.

1) *Trusted authority (TA):* TA is a trusted entity responsible for generating system public parameters. Moreover, TA generates pseudonyms for nodes to enable anonymous interactions and to track the real identities of nodes in the event of malicious behavior. TA is not responsible for generating public-private key pairs for nodes and does not participate in the federated learning training process.
2) *Worker nodes:* As the trainer of FL, the worker node obtains the local model through training, signs it with the private key, and then publishes the local model upload transaction in the blockchain. In order to reduce the communication cost of the blockchain, the local model is stored in the interplanetary file system (IPFS), and the upload transaction contains the storage proof of the local model. A set of workers are referred to as $W = \{w_1, w_2, \ldots, w_n\}, w_i \in W, i = 1, 2, \ldots, n$.
3) *Task publishers:* The task publisher, as the requester of global model training, issues FL training request transactions in the blockchain. Before aggregating the local model, the task publisher first verifies the identity of the workers and ensures that the local model is not tampered with during transmission.
4) *Blockchain:* The proposed DAFL adopts the DAG consortium blockchain framework, and the transaction verification and blocking are conducted by the committee node, which can effectively improve the system throughput. Participants can publish various transactions in the blockchain, including registration transactions, training request transactions, local model upload transactions, etc. The identity information of all nodes is registered in an accumulator jointly maintained by the committee. Only the final limited accumulator value is transmitted on the blockchain, and there is no need to store the identity information of all nodes. When a node joins or exits, the blockchain only needs to update the accumulator value in the latest block, and other nodes can update their witnesses by themselves without any third party. When receiving the transaction, the committee node verifies the illegality of the sender according to the accumulator value in the latest block and the witness of the sender in the transaction. If the verification is established, the transaction will be stored in the cache pool. Otherwise, it will be discarded.

B. Threat Model and Security Requirements

Considering the untrustworthiness of participating entities and the unreliability of network communication, we analyze the potential threats of workers and the insecurity factors in the network. Firstly, workers may upload invalid local models, resulting in slow global model convergence, or even upload

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poisonous models to hinder global model convergence, or workers may deny their uploaded local models. Secondly, there are external attackers in the network, who may hinder the global model convergence by launching various traditional attacks.

Considering the above threats, the proposed DAFL should satisfy the following security requirements.

1) *Authentication and Integrity:* The task publisher can verify the true source of the received local model, and DAFL shall ensure the integrity of the transmitted local model.
2) *Non-repudiation and Unforgeability:* The nodes in the system cannot deny the information they publish, and they cannot forge other nodes to publish information. Similarly, the information released by an attacker who falsifies other legal nodes will also be recognized by the system.
3) *Identity Privacy-Preserving and Traceability:* DAFL should ensure the security of the real identity of the node, so as to avoid the privacy risk caused by identity leakage. Unless TA, no other third party can extract their real identity. When a node has malicious behavior, TA can calculate its anonymous identity to trace its real identity.
4) *Unlinkability:* Unlinkability means that the attacker cannot extract sensitive information from different information of the same entity, That is, there is no association between different interactive behaviors of the same entity.
5) *Resistance to various attacks:* The proposed DAFL scheme should have the ability to withstand multiple known attacks, such as man-in-the-middle attacks and replay attacks, etc.

C. Design Goals

The goal of DAFL is to design a secure and efficient authentication protocol for FL. Firstly, DAFL implements decentralized and anonymous authentication without a trusted third party. Secondly, DAFL must meet the above five security requirements. Finally, in order to guarantee the practical application effect of DAFL, the overhead of the authentication protocol should be lightweight, and the system should achieve high scalability.

V. Federated Learning for Authentication

We propose a decentralized anonymous batch authentication scheme for FL using digital signature technology. For clarity, we specify the relevant notations used in this paper in Table I.

A. Initialization

TA performs the below steps to initialize the FL anonymous authentication scheme and generate system public parameters.

1) TA selects security parameter λ and executes generation algorithm $Gen(\lambda)$ to generate (q, P, G), where G is an additive group with large prime order q, and P is a generator of group G.
2) TA randomly selects $x \in Z_q^*$ and calculates $Y = xP$, where x is the master private key of the system to be kept secret, and Y is the public key of the system.

TABLE I
The Notations and Abbreviations in the Scheme

Notations	Descriptions
G	A cyclic group with order q
P	A generator of G
(x, Y)	The master key pair
(x_i, Y_i)	The private and public key of w_i
PID_i	The pseudo-identity of w_i
TS_i	The validity period of PID_i
T_i	The current timestamp
TA	The Trusted authority
w_i	The worker nodes
$H_1(\cdot), H_2(\cdot), H_3(\cdot)$	One-way hash functions
$\{f_1, f_2, ..., f_m\}$	The committee set
F	The accumulator element set
Acc_F	The accumulator value
$upmsg$	The auxiliary information
wit_i	The witness of element f_i
M_i	The local model trained by w_i
σ_i	The signature of w_i
$\oplus$	The exclusive-OR operation

3) TA chooses three one-way hash functions $H_1 : \{0,1\}^* \times G \to Z_q^*$, $H_2 : \{0,1\}^* \times G \times \{0,1\}^* \times \{0,1\}^* \to Z_q^*$, $H_3 : \{0,1\}^* \to Z_q^*$.
4) Finally, TA publishes the system public parameters $\{q, P, G, Y, H_1, H_2, H_3\}$.

B. Pseudonym Generation

Before joining the DAFL system, TA generates pseudonyms for each node. Nodes participate in FL training through pseudonyms, and TA can recover their real identity for punishment in case of malicious behavior. Take the worker w_i as an example, TA performs the following steps to generate a pseudonym.

1) The w_i sends $\{ID_i\}$ to the TA to request a pseudo identity, where ID_i is the real identity of w_i.
2) TA randomly selects $a_i \in Z_q^*$ and leverages the master private key x to generate a periodic pseudonym $PID_i = ID_i \oplus H_3(xa_i) \oplus H_3(TS_i)$ for w_i, where TS_i is the valid period of the pseudonym. Before TS_i expires, w_i needs to update the pseudo identity to prevent attackers from continuously tracking the same pseudonym and linking to the worker's real identity.
3) Finally, TA returns the temporary pseudo identity $\{PID_i, TS_i\}$ to w_i.

C. Genesis Block Generation

The genesis block is created by the committee of the DAG consortium blockchain, which includes at least m committee member information, namely $\{PID_j, Y_j\}$, $j = 1, \ldots, m$, $m < n$, and the initial accumulator value. The genesis block is generated as follows.

1) First, determine the members of the blockchain committee and generate their respective public-private key pairs $\{x_j, Y_j\}$. Then, the committee nodes initialize the accumulator with security parameters.
2) The committee nodes cooperate to generate the initial accumulator value $Acc_F = u^{f_1 f_2 \cdots f_m} \mod n$, where $u \in$

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Algorithm 1: Identity Addition Function.

Input: Acc_F, F, f_i
Output: Acc_F, $upmsg$
1: **if** $f_i \in F$ **then**
2: return Acc_F
3: **else**
4: $F \leftarrow F \bigcup \{f_i\}$
5: $upmsg \leftarrow f_i$
6: $Acc_F \leftarrow Acc_F^{f_i}$
7: **return** Acc_F, $upmsg$

Z_n, $\{f_1, f_2, \ldots, f_m\} \in F$ is the committee set and $f_j = H_1(PID_j, Y_j)$.

3) A committee puts Acc_F, the committee set $\{f_1, f_2, \ldots, f_m\}$ and timestamp into the genesis block and broadcasts it to all members of the committees.
4) Other committee members verify whether Acc_F is correct. If it is, the genesis block is generated. Otherwise, the block is discarded.

D. Registration

The worker w_i performs the following steps to register with the DAFL system. The registration process for task publishers is the same and will not be described here.

1) The w_i first generates the private key x_i and calculates the corresponding public key $Y_i = x_i P$. The w_i combines pseudonym and public key to calculate $f_i = H_1(PID_i, Y_i)$, randomly selects $e_i \in Z_q^*$, and calculates $E_i = e_i P$ and $\eta_i = x_i H_3(PID_i) + e_i f_i$.
2) The w_i sends registration request transaction $\{PID_i, Y_i, \eta_i, E_i, f_i\}$ to the DAG consortium blockchain to request registration.
3) After the registration request transaction is received, the committee nodes verify whether $\eta_i P = Y_i H_3(PID_i) + f_i E_i$ is valid. If it is, the committee node executes Algorithm 1 to add the identity information $f_i = H_1(PID_i, Y_i)$ of the transaction sender to the accumulator and then obtains the updated accumulator value Acc_F and auxiliary information $upmsg$. In addition, the accumulator set is updated to $F \leftarrow F \bigcup f_i$, thus adding w_i's identity information f_i to the accumulator.
4) Workers generate their own existential witness $wit_i = u^{\prod_{f_j \in F, f_j \neq f_i} F}$ according to the accumulator value. In the FL interaction process, other nodes do not need to traverse the blockchain, and only verify the witnesses wit_i of the workers to judge its legitimacy, thereby improving the verification efficiency.
5) Finally, a committee adds $\{PID_i, Y_i, Acc_F, upmsg\}$ to the latest block. Other committee nodes verify whether Acc_F is correct. If so, record the block in the local blockchain, and update their own witness $wit_i' = wit_i^{upmsg}$ in the accumulator. Otherwise, discard the block.

E. Local Model Signature Generation

The task publisher first publishes the model training request transaction in the blockchain to initiate FL. Then, the w_i leverages private data to train the FL model and obtain the local model M_i. Finally, the w_i signs M_i with its private key x_i, and publishes an update transaction to the blockchain committee node to upload M_i. The w_i generates the signature σ_i for the local model M_i to be transmitted as follows.

1) The w_i randomly selects $r_i \in Z_q^*$ and calculates $R_i = r_i P$.
2) The w_i leverages a pseudonym PID_i and its corresponding public key Y_i to calculate $\delta_i = r_i H_1(PID_i, Y_i)$, and combines the local model M_i to calculate $\alpha_i = H_2(PID_i, Y_i, M_i, T_i)$, where T_i is a current timestamp.
3) The w_i calculates the signature $\sigma_i = x_i \alpha_i + \delta_i$ through the private key x_i and sends the local model upload transaction $\{PID_i, Y_i, M_i, \sigma_i, R_i, wit_i, T_i\}$ to the blockchain.
4) After receiving the local model upload transaction, the committee first extracts $\{PID_i, Y_i\}$ and wit_i from the transaction, then obtains the accumulator value Acc_F from the latest block. Finally, whether $\{PID_i, Y_i\}$ is in the accumulator is verified by judging whether $wit_i^{f_i} = Acc_F$. If it is, the transaction is stored in the latest block.

Considering that transmitting the local model on the blockchain will greatly increase the communication and storage costs, the w_i first stores M_i in IPFS, and the transaction contains the storage hash returned by IPFS.

F. Local Model Verification

After finding the local model upload transaction in the blockchain, the task publisher verifies the identity of the sender and confirms that the local model has not been tampered with during the transmission. After the verification, the task publisher aggregates the global model using local models. The verification steps are as follows.

1) The task publisher first checks whether the T_i is fresh. If not, the local model will be rejected.
2) The task publisher verifies whether the following equation is equal,

$$\sigma_i P = \alpha_i Y_i + R_i H_1(PID_i, Y_i), \tag{1}$$

where $\alpha_i = H_2(PID_i, Y_i, M_i, T_i)$. If the equation is true, the local model is accepted. Otherwise, the local model is rejected.

We show that if equation (1) is satisfied, the uploaded local model is from $\{PID_i, Y_i\}$ and has not been tampered with during transmission. It is known that $\alpha_i = H_2(PID_i, Y_i, M_i, T_i)$, $Y_i = x_i P$, $\delta_i = r_i H_1(PID_i, Y_i)$, and

$$\begin{aligned} \sigma_i P &= (\alpha_i x_i + \delta_i) P \\ &= (\alpha_i x_i) P + \delta_i P \\ &= \alpha_i Y_i + R_i H_1(PID_i, Y_i). \end{aligned} \tag{2}$$

Therefore, the local model can be verified correctly.

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The proposed authentication scheme supports batch verification, verifying multiple signatures at one time to reduce the verification overhead. In order to prevent attackers from attacking aggregation signatures, the task publisher leverages the small coefficient test technology [38] to verify the validity of these signatures through the following formula, where $l_i \in [1, 2^d]$ and l_i is a small integer.

$$\left(\sum_{i=1}^{n} l_i \sigma_i\right) P = \sum_{i=1}^{n} l_i \alpha_i Y_i + \sum_{i=1}^{n} l_i R_i H_1(PID_i, Y_i). \quad (3)$$

All signatures are valid and acceptable if the above equation holds; Otherwise, at least one signature is invalid, and the task publisher can run equation (1) for each signature to check it out. We analyze that if the batch signatures are correct and untampered, then they must satisfy equation (3). Due to $\alpha_i = H_2(PID_i, Y_i, M_i, T_i)$, $Y_i = x_i P$, $\delta_i = r_i H_1(PID_i, Y_i)$, and $R_i = r_i P$, we obtain

$$\begin{aligned}\left(\sum_{i=1}^{n} l_i \sigma_i\right) P &= \left(\sum_{i=1}^{n} l_i \left(x_i \alpha_i + \delta_i\right)\right) P \\ &= \left(\sum_{i=1}^{n} l_i x_i \alpha_i\right) P + \left(\sum_{i=1}^{n} l_i \delta_i\right) P \\ &= \sum_{i=1}^{n} l_i \alpha_i Y_i + \left(\sum_{i=1}^{n} l_i r_i H_1(PID_i, Y_i)\right) P \\ &= \sum_{i=1}^{n} l_i \alpha_i Y_i + \sum_{i=1}^{n} l_i R_i H_1(PID_i, Y_i). \quad (4)\end{aligned}$$

Therefore, multiple local models can be correctly verified.

If equation (3) does not hold, and the task publisher finds some illegal messages, it sends the pseudonym PID_i of the illegal message owner to TA. Then, TA calculates $ID_i = PID_i \oplus H_3(xa_i) \oplus H_3(TS_i)$ to restore the real identity ID_i of the illegal message owner and punishes it.

G. Identity Revocation

When some pseudonyms are about to expire or the malicious times reach the threshold, the committee node will delete the corresponding pseudonym information $\{PID_i, Y_i\}$ from the accumulator, as shown in Algorithm 2 and add the updated accumulator value Acc'_F to the latest block. Other block receivers verify whether Acc'_F is correct. If so, add the block to the local blockchain and update their own witness. Otherwise, the block is discarded.

Algorithm 2: Identity Revocation Function.

Input: Acc_F, F, f_i
Output: Acc'_F, $upmsg$
1: **if** $f_i \notin F$ **then**
2: return Acc_F
3: **else**
4: $F \leftarrow F \setminus \{f_i\}$
5: $Acc'_F \leftarrow u^{\prod_{f_i \in F} F}$
6: $upmsg \leftarrow \{f_i, Acc_F, Acc'_F\}$
7: **return** Acc'_F, $upmsg$

VI. Security Proof and Analysis

We first prove the security of our proposed authentication algorithm. Then, according to the security requirements, DAFL is analyzed in detail.

A. Security Proof

In the random oracle model, we demonstrate that the proposed authentication algorithm is unforgeable under the assumption that the DL problem is hard. The detailed proof is as follows.

Definition 4: The unforgeability of the authentication scheme $(KeyGen, Sign, Verify)$ under the chosen message attack is defined by the game between the challenger $\mathcal{B}$ and the adversary $\mathcal{A}$.

Setup: Challenger $\mathcal{B}$ executes $KeyGen$ to generate public parameters, randomly selects the master private key and keeps it secretly. Then, the system public parameters are sent to the adversary $\mathcal{A}$.

Hash Query: The adversary $\mathcal{A}$ can initiate a hash oracle query to all the hashes used in the scheme at any time, and the challenger $\mathcal{B}$ returns the corresponding hash.

Signing Query: The adversary $\mathcal{A}$ freely chooses the messages $M_1, \ldots, M_{q_s} \in \{0,1\}^*$ for signature query, and asks at most q_s times. The challenger responds with signature $\sigma_i = Sign(SK_i, M_i)$ to each query.

Forgery: The adversary $\mathcal{A}$ outputs a pair (M, σ). If M does not belong to any of $M_1, \ldots, M_{q_s}$, and σ is a valid signature of the message M, adversary $\mathcal{A}$ wins the game.

Definition 5: In the existential unforgeability against chosen-message attacks (EU-CMA) security model, a signature scheme is $(t, q_{H_1}, q_{H_2}, q_s, \varepsilon)$-secure if $\mathcal{A}$ cannot win the above game with an advantage ε within time t after q_H queries for hash function and q_s signature queries. The ε is the probability that adversary $\mathcal{A}$ successfully forges the signature.

Theorem 1: Let G be a (t', ε')-DL group of order q. Then the signature algorithm based on G_1 has $(t, q_{H_1}, q_{H_2}, q_s, \varepsilon)$-secure against the EU-CMA in the random oracle model.

Proof: Suppose $\mathcal{A}$ is the forgery algorithm that can compromise a signature algorithm with $(t, q_{H_1}, q_{H_2}, q_s, \varepsilon)$. We will show how to institute an algorithm that can solve the DL problem on group G with probability ε', and the running time is at most t'. Let P be the generator of the group G, and the algorithm $\mathcal{B}$ has known the parameters of the DL problem aP, and the purpose is to output a. The algorithm $\mathcal{B}$ simulates the interaction between the challenger and the adversary $\mathcal{A}$ as follows.

Setup: Challenger $\mathcal{B}$ sends generator P, public key $Y_i = aP$ and the public parameters to adversary $\mathcal{A}$, where private key x_i is equivalent to a.

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Adversary $\mathcal{A}$ adaptively makes the following oracle queries. Before receiving a query from adversary $\mathcal{A}$, challenger $\mathcal{B}$ generates two initially empty hash lists and one initially empty signature list to record all queries and responses, where the lists are initially empty.

H_1 query: The adversary $\mathcal{A}$ makes a hash lookup at this phase. The $\mathcal{B}$ maintains an H_1 list $L_1 = (PID_i, Y_i, h_{1i})$. When $\mathcal{B}$ receives an H_1 query, B first checks the H_1 list. If there are relevant query records, it directly returns the corresponding h_{1i}. Otherwise, $\mathcal{B}$ selects the random number $h_{1i} \in Z_q^*$, makes $h_{1i} = H_1(PID_i, Y_i)$, returns h_{1i} to $\mathcal{A}$, and adds (PID_i, Y_i, h_{1i}) to the list L_1.

H_2 query: If adversary $\mathcal{A}$ issues a random oracle query H_2, $\mathcal{B}$ looks up the hash list $L_2 = (PID_i, Y_i, M_i, T_i, h_{2i})$ to get the corresponding answer. If the tuple $(PID_i, Y_i, M_i, T_i, h_{2i})$ exists in hash list, then $\mathcal{B}$ answers with h_{2i}. Otherwise, $\mathcal{B}$ generates h_{2i} uniformly at random from Z_q^*, answers with it, and adds the tuple $(PID_i, Y_i, M_i, T_i, h_{2i})$ to the L_2 list.

Sign query: If the adversary $\mathcal{A}$ issues a signature query (PID_i, M_i) where $1 \le i \le q_s$, $\mathcal{B}$ looks up the signature list L_S to get the corresponding answer. If a tuple (PID_i, M_i, σ_i) exists in the L_S, $\mathcal{B}$ answers with (PID_i, M_i, σ_i). When (PID_i, M_i) is a new signature query, $\mathcal{B}$ first performs H_1, H_2 queries to recover tuples (PID_i, Y_i, h_{1i}) and $(PID_i, Y_i, M_i, T_i, h_{2i})$ from lists L_1, L_2, respectively. Then, $\mathcal{B}$ randomly selects $\sigma_i \in Z_q^*$, let $Y_i = \alpha_i^{-1}(\sigma_i P - \beta_i R_i)$. Since $\sigma_i P = \alpha_i Y_i + \beta_i R_i$ holds, the above signature σ_i is a valid signature.

Forgery: After a series of queries mentioned above, the adversary $\mathcal{A}$ finally generates a signature σ^* about (PID^*, M^*), and ensures that the (PID^*, M^*) has not been queried in the previous hash and signature query. According to AGM definition, σ^* is calculated from the group elements given by $\mathcal{B}$, and $\mathcal{B}$ knows all the calculation methods of its own group elements. Therefore, $\mathcal{B}$ can know α_i and δ_i, satisfying $\sigma^* = a\alpha_i + \delta_i$. Let α_i be any integer, $\delta_i = 0$, then $\sigma^* = a\alpha_i + \delta_i = a\alpha_i$, and $a = \sigma^*/\alpha_i$ can be obtained, that is, $\mathcal{B}$ solves an instance a of DL problem.

B. Security Analysis

We analyze that DAFL satisfies various security requirements presented in Section IV.

1) Authentication and Integrity: In the DAFL scheme, all local models are signed by workers before being sent to the Blockchain committee. As stated in Theorem 1, the proposed signature scheme can resist the EU-CMA in the random oracle model. Worker w_i leverages the private key x_i to sign the local model, and if the task publisher verification $\{PID_i, Y_i, M_i, \sigma_i, R_i, wit_i, T_i\}$ passes, the source authentication is implemented. This also ensures the integrity of the local model, since any bit change will result in authentication failure.

2) Non-Repudiation and Unforgeability: During FL interaction, all local models are signed by workers with private keys. The signature algorithm is based on the DL assumption, and the attacker cannot obtain the private key x_i by compromising the signature σ_i. When equations $\sigma_i P = \alpha_i Y_i + R_i H_1(PID_i, Y_i)$ and $(\sum_{i=1}^n l_i \sigma_i)P = \sum_{i=1}^n l_i \alpha_i Y_i + \sum_{i=1}^n l_i R_i H_1(PID_i, Y_i)$ are established, the worker cannot deny the local model it sends. Also, since there is no private key, an attacker cannot forge the local model.

3) Identity privacy-Preserving and Traceability: Before FL training, TA generates pseudonyms $\{PID_i, TS_i\}$ for nodes, and all nodes use pseudonyms PID_i to send and receive transactions. The pseudonym $PID_i = ID_i \oplus H_3(xa_i) \oplus H_3(TS_i)$ is calculated by the TA's master private key x. In order to obtain the real identity of the node, the adversary has to obtain the master private key x and the a_i, which is obviously impossible. Therefore, the proposed scheme can ensure the identity privacy of nodes. As mentioned above, since the $\{PID_i, TS_i\}$ is generated by TA, it can also leverage x to recover the real identity ID_i of the node when illegal behavior occurs, that is, calculate $ID_i = PID_i \oplus H_3(xa_i) \oplus H_3(TS_i)$ to realize the tracing of the node.

4) Unlinkability: Each pseudonym PID_i has a validity period TS_i. When the validity period expires, nodes will leverage the new pseudonym PID_i' to interact. To link the two pseudonyms $PID_i = ID_i \oplus H_3(xa_i) \oplus H_3(TS_i)$ and $PID_i' = ID_i \oplus H_3(xa_i') \oplus H_3(TS_i')$, the adversary has to acquire x, a_i, and a_i' at the same time, which is impossible. Therefore, our DAFL scheme can guarantee the un-linkability of multiple pseudonyms from the same source.

5) Resistance to Traditional Attacks: According to Theorem 1, it is not possible to impersonate a w_i and generate tuples $\{PID_i, Y_i, M_i, \sigma_i, R_i, wit_i, T_i\}$ of the local model M_i with valid signatures σ_i. Therefore, DAFL can prevent man-in-the-middle attacks. Each transaction in the blockchain contains a timestamp T_i, and nodes can resist replay attacks by verifying the timestamp T_i.

VII. Comparisons and Performance Analysis

We begin with a comprehensive comparison of DAFL with related authentication schemes. Then, extensive experiments are performed to demonstrate the performance of DAFL.

A. Comprehensive Comparisons of DAFL

We compare DAFL with Refs. [11], [14], [31], [32], as shown in Table II. We mainly discuss the schemes from the aspects of decentralization, defense of man-in-the-middle attacks, anonymity, traceability, lightweight, and whether it supports fast query. The authentication schemes in works [11], [14], [31] and [32] all require the third-party KGC to generate partial private keys for the nodes according to their identity information for authentication. Therefore, they are centralized, with a single point of failure, key management, and identity storage problems. Although all schemes can resist man-in-the-middle attacks, Refs. [14] and [31] leak the identity privacy of participants. In order to hide the real identity of participants, Refs. [11], [32] and DAFL all implement anonymous interaction of participants. However, in Ref. [11], it is difficult to track and punish malicious behavior. On the other hand, all these four comparison schemes implement identity authentication through bilinear pairing. Considering the high computational complexity of bilinear pairing, these schemes are computationally expensive. Moreover, Refs. [31] and [32] use blockchains to record

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TABLE II
COMPARISON OF RELATED AUTHENTICATION SCHEMES

Scheme	Decentralization	Man-in-the-middle	Anonymity	Traceability	Lightweight	Fast query
Ref. [11]	×	√	√	×	×	-
Ref. [14]	×	√	×	√	×	-
Ref. [31]	×	√	×	√	×	×
Ref. [32]	×	√	√	√	×	×
DAFL	√	√	√	√	√	√

-: no such operation in the literature.

TABLE III
SUMMARY OF TIME CONSUMPTIONS FOR CRYPTOGRAPHIC CALCULATION OPERATIONS

Notations	Descriptions	Time(ms)
T_p	A bilinear pairing operation	4.452
T_e	An exponential operation	0.202
T_m	A scale multiplication operation	0.071
T_h	An one-way hash operation	0.083
T_a	A point addition operation	0.041

TABLE IV
COMPARISON OF COMPUTATIONAL TIME COST WITH BENCHMARKS

Scheme	Signature Time (ms)	Verification Time of a message (ms)	Verification Time of n messages (ms)
Ref. [14]	$T_e + T_h \approx 0.285$	$2T_m + 3T_h + T_p + T_e \approx 5.045$	$2nT_m + 3nT_h + nT_p + nT_e \approx 5.045n$
Ref. [31]	$T_e + T_p + T_m + T_h \approx 4.808$	$2T_p + T_m + T_h \approx 9.058$	$2nT_p + nT_m + nT_h \approx 9.058n$
DAFL	$T_m + 2T_h \approx 0.237$	$3T_m + T_a + 2T_h \approx 0.42$	$(2n+1)T_m + T_a + 2nT_h \approx 0.071(2n+1) + 0.166n$

public key or pseudonym information. Due to the lack of an effective blockchain query mechanism, they can only query the public key or pseudonym by traversing the blockchain. As the number of transactions increases, information queries will become more and more time-consuming. To sum up, these identity authentication schemes have severe shortcomings and are therefore not suitable for large-scale FL authentication.

We propose a decentralized DAFL based on blockchain and accumulator, which does not require a trusted third party to generate keys for nodes. Nodes generate their own public-private key pair and add $\{PID_i, Y_i\}$ to the accumulator, which is maintained by the blockchain committee node. Therefore, decentralized FL node registration and authentication are achieved. Although there is a TA in our scheme, it does not participate in the FL authentication process. DAFL realizes the anonymous training of participating nodes and reveals their true identities in the event of malicious behavior. In addition, our proposed signature algorithm does not involve bilinear pairing and exponential operation, and it supports batch verification to achieve lightweight FL authentication. When querying the pseudonym and public key $\{PID_i, Y_i\}$ on the DAG blockchain, it only needs to check the acc_F in the latest block, which greatly improves the query efficiency of the blockchain.

Moreover, Refs. [34] and [37] have given detailed evaluations of the DAG blockchain framework and accumulator, respectively, and it is clear that they have great potential in improving the scalability of the blockchain system. Therefore, this paper does not verify the performance advantages of the DAG consortium blockchain architecture and accumulator. In general, compared with existing authentication schemes, our DAFL has many functional advantages and is more suitable for FL identity authentication in a distributed network environment.

B. Computation Cost Comparisons

This section demonstrates the computational overhead of DAFL and compares it with Refs. [14] and [31] under the same experimental configuration. Table III lists various cryptographic computation operations and their execution times. The execution time of cryptographic operations was obtained using the public pairing-based cryptography (PBC) library [39] on a laptop with an Intel i5-2450 M processor (8 Gigabytes RAM) on Windows 7 operating system.

Table IV shows the computational cost of the three schemes at different stages. When calculating a message M_i's signature, Ref. [14] involves one exponential T_e and a hash T_h operations, Ref. [31] requires a bilinear pairing T_p, an exponential T_e, a multiplication T_m on G and a hash T_h operations, while our scheme only require a multiplication T_m and two hash T_h operations. To verify a single signature, Ref. [14] includes a bilinear pairing T_p operation, an exponential T_e operation, two multiplication T_m operations, and three hash T_h operations; Ref. [31] includes two bilinear pairing T_p operations, one multiplicative T_m operation, and one hash T_h operation. Instead, our scheme requires only three multiplicative T_m operations, one addition T_a operation, and two hash T_h operations. In order to compare the computational cost of Refs. [14], [31] and our scheme more clearly, we describe the computational cost in terms of computing a signature and implementing an identity authentication, as shown in Fig. 1. It can be seen that when calculating a signature, DAFL has the lowest cost, and the computational cost of Ref. [31] is much higher than that of DAFL and Ref. [14]. When verifying a signature, Ref. [31] has the largest cost because there are two T_p operations, followed by Ref. [14], and DAFL is the smallest. From Fig. 2, DAFL has lower computational overhead in terms of signature and single verification.

To demonstrate the advantage of DAFL in batch identity authentication, we compare the calculation cost of batch authentication for n messages in our scheme with the Refs. [14] and [31] in Fig. 3. As illustrated, the batch verification cost of our scheme and Refs. [14], [31] increases linearly with the number of authentication messages. When authenticating n signatures in batches, DAFL requires $(2n+1)$ multiplication operations on G, a point addition T_a operation, and $2n$ hash operations. However, in Refs. [14] and [31], the bilinear pairing T_p operation increases with the number of messages during batch validation, so the growth rate of Refs. [14] and [31] is much higher than our

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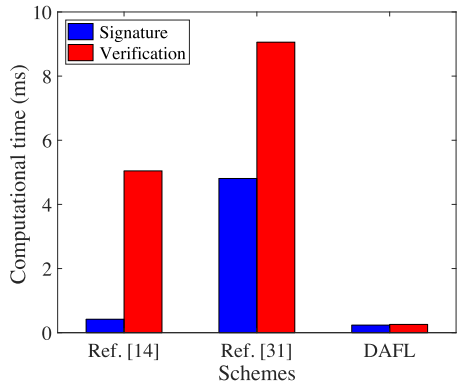


Fig. 2. Comparison of computation cost for signing and verifying a message.

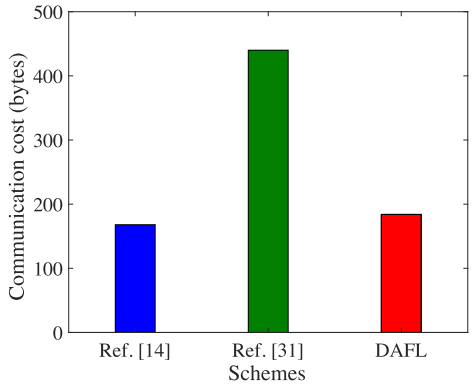


Fig. 4. Comparison on communication cost of three schemes.

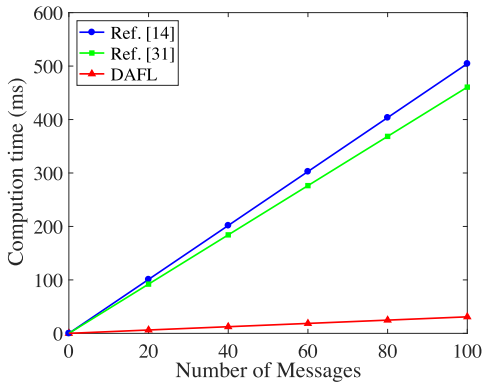


Fig. 3. Comparison of computation cost for batch verification.

scheme. When 100 messages are verified in batch, our scheme only requires 31 ms of computation cost, while Refs. [14] and [31] require 504 ms and 460 ms, respectively. To sum up, compared with relevant authentication schemes, DAFL has a significantly lower computational cost, and as the number of messages increases, the advantages of our scheme become more obvious. Therefore, our scheme can achieve more lightweight and scalable FL identity authentication.

C. Communication Cost Comparisons

We discuss the communication cost of DAFL sending an authentication message to the committee node. Furthermore, we compared the communication cost of DAFL with that of Refs. [14] and [31] under the same settings. Consider that the primes p and $\hat{p}$ are 20 and 64 bytes respectively, then the additive group G and the multiplicative group G_T produce 40 and 128 bytes of output, respectively. For uniformity, it is assumed that worker w_i's identity PID_i, hash function's output, a random nonce, the message M_i, and timestamp T_i are 20 bytes, 20 bytes, 20 bytes, 20 bytes, and 4 bytes, respectively.

In Ref. [14], in order to send a local model parameter M_i, worker w_i broadcasts the tuple $\{\sigma_i, Y_i, M_i, ID_i, ID_j, T_i\}$, where $\sigma_i = (h_i, S_i, T_i)$ is the signature of M_i, h_i is the output of the hash function, $\{S_i, Y_i\} \in G$, ID_i and ID_j are the identities of the sender and receiver and T_i is the timestamp. Accordingly, the communication overhead of Ref. [14] is $20 \times 4 + 40 \times 2 + 4 \times 2 = 168$ bytes. In Ref. [31], the tuple $\{\sigma_i, M_i, ID_i, ID_j, Y_i, T_i\}$ is broadcast, where $\sigma_i = (V_i, \sigma_{i1}, \sigma_{i2})$ is the signature, $\{V_i, ID_i, ID_j, Y_i\} \in G$ and $\{\sigma_{i1}, \sigma_{i2}\} \in G_T$. Thus, the communication cost of Ref. [31] is $40 \times 4 + 128 \times 2 + 20 + 4 = 440$ bytes. In our scheme, it takes $20 \times 5 + 40 \times 2 + 4 = 184$ bytes for a worker w_i to send a tuple $\{PID_i, Y_i, M_i, \sigma_i, R_i, wit_i, T_i\}$ to a blockchain committee node, where $\{\sigma_i, wit_i\} \in Z_q^*$ and $\{Y_i, R_i\} \in G$. From Fig. 4, in terms of the communication overhead of sending an authentication message, the overhead of DAFL is almost equivalent to Ref. [14] and significantly lower than Ref. [31]. Therefore, our scheme has lower communication overhead.

Furthermore, both our proposed scheme and Ref. [31] utilize blockchain for authentication. The above analysis does not consider the communication overhead of blockchain consensus. Our scheme only submits local model upload transactions to a committee node, while Ref. [31] broadcasts transactions across the blockchain network. Obviously, DAFL has advantages in communication overhead.

D. FL Performance Analysis

We leverage the Fashion-MNIST and CIFAR10 datasets as in [40] to perform experiments on the PyTorch platform to show the effectiveness of DAFL compared with the traditional attack-free FL systems and attacking FL systems (FL-ATT). The traditional attack-free FL systems mean that local models are honestly trained by workers. All the experiments are implemented on an Ubuntu 16.04 server with 62 GB RAM, 48 CPU cores and two NVIDIA GeForce GTX 1080 Ti GPUs. The Fashion-MNIST and CIFAR10 datasets are split into 100 shards and assigned to 100 workers, from which 50 workers are selected at random in each round to perform global model training. Stochastic gradient descend (SGD) is computed locally on each dataset, the learning rate is 0.02, the batch size is 32, and workers are trained locally for 5 epochs each time. A lightweight convolutional neural network (CNN) model is adopted as the training model. The detailed model structure configuration is shown in Table V. For comparison, we set some bad workers who cannot be authenticated because of insufficient bandwidth or illegal identity. Bad workers will upload low-quality local models, which are derived by adding Gaussian noise. We assume that local models uploaded by authenticated workers are honestly trained with local data.

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TABLE V
CNN MODEL STRUCTURE

Dataset	Layer	Number of filters	Size of filter	Size of output
Fashion-MNIST	Input	-	-	28×28×1
	Convolution	32	5×5	32×24×24
	Pooling	32	2×2	32×12×12
	Convolution	64	5×5	64×8×8
	Pooling	64	2×2	64×4×4
	Linear	-	-	512
	Linear	-	-	10
CIFAR10	Convolution	32	3×3	32×32×32
	Convolution	64	3×3	64×32×32
	Pooling	64	2×2	64×16×16
	Convolution	128	3×3	128×16×16
	Convolution	128	3×3	128×16×16
	Pooling	128	2×2	128×8×8
	Convolution	256	3×3	256×8×8
	Convolution	256	3×3	256×8×8
	Pooling	256	2×2	256×4×4
	Linear	-	-	1024
	Linear	-	-	512
	Linear	-	-	10

TABLE VI
PERFORMANCE COMPARISON UNDER DIFFERENT PROPORTIONS OF BAD WORKERS

Bad worker	Scheme	DataSets	
		Fashion-MNIST	CIFAR 10
10%	FL	0.8935	0.8150
	DAFL	0.8929	0.8141
	FL-ATT	0.8556	0.4732
15%	FL	0.8921	0.8180
	DAFL	0.8904	0.8113
	FL-ATT	0.8371	0.4071
20%	FL	0.8903	0.8164
	DAFL	0.8888	0.8109
	FL-ATT	0.8267	0.3365
25%	FL	0.8923	0.8170
	DAFL	0.8881	0.8092
	FL-ATT	0.8254	0.1011

We conduct multiple sets of experiments to verify that our scheme has good accuracy under different proportions of bad workers when the standard deviation of noise $\xi = 0.3$. In the following experiments the number of communication rounds is 60. As shown in Table VI, when there are 10%, 15%, 20%, and 25% bad workers, the accuracy of our scheme on both Fashion-MNIST and CIFAR10 datasets is infinitely close to the attack-free FL scheme. This is because in the DAFL scheme bad workers will fail authentication and their low-quality local models will not be adopted. After randomly selecting 50 workers, DAFL will first verify them with the proposed identity authentication algorithm, and the local models that fail the authentication will be deleted. Then, reselect from the remaining workers until authentication is established. The accuracy of the FL-ATT scheme on both datasets decreases with the increase of bad workers. When there are 25% bad workers, the accuracy of the FL-ATT scheme on the two datasets is only 82.5% and 10.1%, respectively.

In addition, we also analyze the effects of different ξ on the accuracy of the three schemes on the two datasets. As shown in Table VII, when there are 25% bad workers, our scheme and the FL scheme still have very close high accuracy no matter how ξ is changed. Since FL is attack-free, all workers train the local model honestly, and our scheme is able to identify bad workers. When ξ increases from 0.1 to 0.5, the accuracy of FL-ATT on the Fashion-MNIST dataset decreases by 14.54%. When $\xi > 0.3$, the model accuracy of the FL-ATT scheme on CIFAR10 remains at 10%, and the model is no longer available. Therefore, we use $\xi = 0.3$ in the above experiments.

TABLE VII
COMPARISON OF DIFFERENT ξ

Noise	Scheme	DataSets	
		Fashion-MNIST	CIFAR 10
ξ=0.1	FL	0.8911	0.8163
	DAFL	0.8886	0.8109
	FL-ATT	0.8811	0.6929
ξ=0.2	FL	0.8904	0.8157
	DAFL	0.8885	0.8060
	FL-ATT	0.8583	0.4704
ξ=0.3	FL	0.8923	0.8170
	DAFL	0.8881	0.8092
	FL-ATT	0.8254	0.1011
ξ=0.4	FL	0.8935	0.8143
	DAFL	0.8878	0.8077
	FL-ATT	0.7712	0.1000
ξ=0.5	FL	0.8887	0.8150
	DAFL	0.8875	0.8065
	FL-ATT	0.7357	0.1000

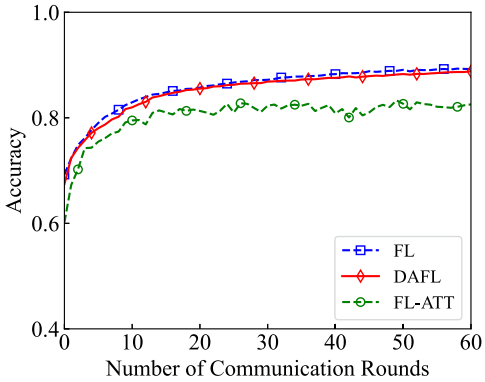


Fig. 5. Accuracy curves with 25% bad workers on Fashion-MNIST.

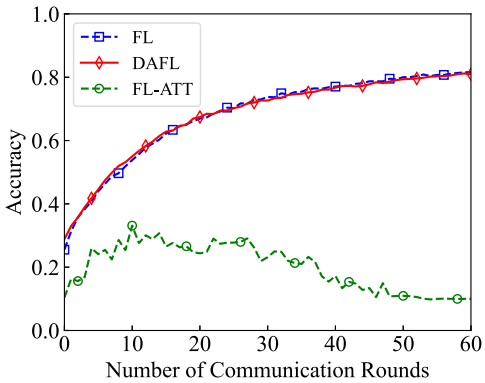


Fig. 6. Accuracy curves with 25% bad workers on CIFAR10.

For a more intuitive display, we show the accuracy curves of the three schemes when there are 25% bad workers. From Figs. 5 and 6, the global accuracy of our DAFL scheme and the FL scheme is almost the same after 60 rounds of iterations on both two datasets. This is due to the fact that our scheme will authenticate workers, and thus bad workers will be identified and

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eliminated. Compared with FL-ATT, DAFL improves the accuracy by 6.27% and 70.81% on Fashion-MNIST and CIFAR10 datasets, respectively.

VIII. Conclusion

In this paper, we first propose a decentralized FL authentication framework, which can achieve anonymous and traceable worker authentication. Then, a computationally efficient batch verification algorithm is proposed, where task publishers are able to batch verify local models of multiple workers, thereby reducing the computational burden in the FL identity authentication process. Formal security proof and analysis show that, in addition to resisting traditional attacks, DAFL can also meet various security requirements of identity authentication. Comprehensive comparisons show that DAFL is superior to the existing scheme in function. Additionally, the performance experiments demonstrate that DAFL has a lower overhead compared with existing authentication schemes. Finally, compared to traditional FL, DAFL has higher global model accuracy in the presence of malicious workers. In the future, we will study how to achieve privacy-preserving and efficient FL when authenticated workers provide inefficient or even malicious local models.

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